

Delay Analysis of Infrastructure Construction Projects in India

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Abstract This study investigates the causes and mitigation strategies for delays in infrastructure construction projects in India, with a focus on roads and bridges. Through a comprehensive methodology involving a questionnaire survey of 72 construction management professionals and the application of the Relative Importance Index (RII), the research identifies material-related issues (RII = 0.562), construction site challenges (RII = 0.555), and contractor-related inefficiencies (RII = 0.506) as the primary causes of delays. Key factors include late material deliveries, land acquisition hurdles, design errors, and delayed payments. Sector-specific analysis reveals land acquisition as critical for road projects (RII = 0.68) and design errors for bridges (RII = 0.55). Factor analysis further categorizes delays into four dimensions: Resource & Supply Chain, Government & External Factors, Contractor Management, and Design & Planning. The study proposes targeted mitigation strategies, such as securing reliable suppliers, proactive stakeholder engagement, and advanced planning tools like BIM. Emphasizing the importance of time management alongside cost and quality, the findings provide actionable insights for stakeholders to enhance project efficiency and reduce delays in India's infrastructure sector.

Keywords Construction delay · Relative importance index (RII) · Sector-specific analysis · Road · Factor analysis · Time

Introduction

Indian construction is a crucial component of an economy that remains highly fragmented. The infrastructure sector is a key driver of the Indian economy, encompassing power, bridges, dams, roads, and urban development. This sector is vital for India's overall development and has garnered significant attention from the government, which is keen on implementing policies to ensure the timely creation of world-class infrastructure. Globally, the construction industry accounts for 6–7% of many countries' gross domestic product, with the total value of annual construction projects exceeding one trillion dollars. Despite its significance, the construction sector frequently encounters project delays, which may arise from various sources including the employer, the contractor, or third parties. These delays not only result in substantial financial losses but also have adverse effects on society.

Numerous studies on the construction industry worldwide have identified time overruns as a common issue, often using "time overrun" and "delay" interchangeably. [1] conducted research on the causes of delays and inefficiencies in Nigeria's construction industry, identifying material scarcity, delayed payments, and poor contract management as primary reasons for slowdowns. In the United States, [2] found that the main reasons for construction delays included labor shortages, weather conditions, and issues with subcontractors. Similarly, a study by Arditi et al. [3] examined delays in Turkish government projects, pinpointing resource shortages, financial difficulties of public organizations and contractors, organizational deficiencies, frequent job delays, widespread change orders, and significantly increased costs as major causes. Chan and [4] investigated the factors causing delays in Hong Kong's construction projects. Their study highlighted site oversight, low selection rates across all project teams, client variations, and essential work revisions

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as the five most common reasons for delays. In Lebanon, [5] identified economic issues, contractual relationships, and project management problems as the main causes of delays in the construction industry. Research in Kuwait by [6] pointed to slow financial procedures and late payments as significant delay factors. Additionally, [7] studied construction project slowdowns in Ghana, contributing further to the understanding of delay causes in the global construction industry.

[8] conducted a study on the causes of delays in Qatar's infrastructure projects, finding that over 80% of these projects experienced delays, with an average delay of 25%. Key factors contributing to these delays included slow response times by utility agencies, significant design changes during construction, ineffective progress management, and frequent modifications to project scope. Another study by [9] focused on delays in road construction infrastructure projects, identifying material shortages and payment issues between contractors as the main reasons for delays. In Kenya, [10] investigated the causes of delays in road construction projects. The study concluded that improvements in financial management structures are necessary for clients to provide timely payments to contractors, which would help mitigate delays.

For delays in Indian construction projects, [11] conducted a survey to analyze the key elements influencing delays in project execution. The study identified the most critical factors as lack of commitment, poor site coordination, inadequate communication, and substandard contractual agreements. Venkatesh et al. [12] and Narayana S [13] also examined the causes of delays in Indian construction projects, focusing on private, government, and non-governmental organizations. Venkatesh [12] study provided recommendations for improved project management techniques and processes that can be implemented during the project's conceptual and detailed scheduling phases to minimize construction delays.

Many research papers have focused on general construction projects, with only a few studies in India addressing delays in infrastructure construction projects. These papers have identified various patterns of delay causes. Most studies have used surveys to pinpoint the key components contributing to construction delays. Different methods, such as the Relative Importance Index (RII), Average Index, and Mean Index, have been employed to determine these causes. The significant parameters causing delays, as identified by most studies, include delayed payments by the owner, contractor experience, design modifications by the owner, poor site management and oversight by the contractor, and change orders by the owner. There is limited research specifically focused on delays in infrastructure construction projects in India. This study aims to investigate the significant factors causing delays in the construction of infrastructure projects.

The objectives of this study are to identify the factors affecting delays in infrastructure construction projects, rank these factors, and suggest delay mitigation plan. The scope of this work is limited to roads and bridges.

Research Methodology

The methodology for this study on delay analysis in Indian infrastructure construction projects was designed to systematically identify, analyze, and rank the causes of delays, with a focus on roads and bridges. The research followed a structured approach, combining preliminary surveys, questionnaire-based data collection, and advanced analytical techniques. Below is a detailed breakdown of the methodology:

The preliminary survey focused on road construction projects primarily managed by the Public Works Department. This survey was crucial as it provided precise information necessary for project planning, including estimating quantities, time schedules, and costs. The data collected from 50 road construction projects included the project name, location, estimated cost, start date, estimated completion date, and the actual delay (in months). The survey revealed that

Table 1 Factors influencing project delay

Sl.No	Factors affecting delay of the project
1	Lack of commitment
2	Insufficient site management
3	Poor site condition
4	Improper planning
5	Lack of clarity in project scope
6	Sub-standard contract
7	Delayed payments
8	Contract modifications
9	Economic problems
10	Restricted access at site
11	Low quality of construction materials
12	Issues regarding permissions approval from stakeholders
13	Obtaining permits from government
14	Poor means of contracting
15	Building codes using in design of projects
16	Major accidents
17	Late land handover by owner
18	Poor site layout
19	Delay in commencement
20	Inappropriate construction methods
21	Transportation delays
22	Improper or wrong cost estimation
23	Rework because of errors during construction
24	Complexity of project
25	Duration of project

almost all projects experienced delays ranging from 1 to 12 months. Additionally, various factors affecting delays were identified from the literature and presented in Table 1.

Following the identification of delay factors from the literature, a questionnaire template was devised to assess these factors. Time overrun is a prevalent issue in the construction industry, prompting an examination of factors influencing project timelines. Respondents were asked to rate these factors using Likert's scale (1–5), with provision for offering suggestions to enhance the study. The questionnaire revealed that the majority of respondents were clients (41.33%), followed by consultants (38.90%) and contractors (12.50%). Regarding the types of projects experienced by respondents, the breakdown was as follows: Buildings (31.3%), Roads (33.6%), Underpass/Bridges (21.40%), Utilities (8.41%), and Others (5.33%). Of the ongoing projects, 77.80% of respondents reported experiencing delays. Concerning the average delay time of delayed projects, 56% of respondents indicated delays of 10 to 30%, while 25% reported delays of 31 to 50%, and 19% cited delays of 51 to 100%.

Data Analysis

The fishbone diagram, depicted in Fig. 1, serves as a tool to visually represent specific causes of delay. Feasible reasons are illustrated on linked branches, each representing different components. As the branches extend outward, with the phase of detail engineering serving as a central point, external branches emerge, connecting to internal branches. The outermost branches depict the root causes of the issue in general. Construction management professionals were tasked with evaluating delay factors based on their

professional judgment and expertise. Data analysis relied on the Relative Significance Index (RSI) technique, which aids in determining the relative significance of reasons for delays. RII values are calculated using Eq. (1) for each variable.

$$RII = \sum W / (A * .N) \quad (1)$$

where RII = Index method of relative significance, .W = weighting of each respondent as allocated on Likert's scale ranges from 1 to 5, A = maximum weight (i.e., 5 in this case), N = complete number of members. RII for variables influencing project time the answers of the participants are calculated. For example consider the factor “planning and scheduling deficiency” Formula for RII = $\sum W / (A * N)$.

$\sum W$ is the sum of all weightages given by all 72 respondents.

H is the largest accessible ranking of 5 in this situation and N is the complete amount of participants who responded the questions, which turns out to be 360 (5*72). Therefore $RII = 200/360 = 0.55$. The same procedure can be used for calculating RII factors.

The study utilized the RII method to analyze other factors, and the results are presented in Table 2. Based on the rankings displayed in Table 2, the mean RIIs for all groups are summarized in Table 3. Upon examining the results presented in Table 3, it is evident that material-related issues are the most frequent cause of delay, securing the top rank. Following closely, construction site-related factors are ranked second, with contractor-related issues in the third position. Project-related causes of delay are the least frequent, occupying the 8th position in the rankings.

The sector-specific analysis reveals important variations in delay causation patterns across different project types. The

Fig. 1 Fish bone diagram for factors which causes schedule delays

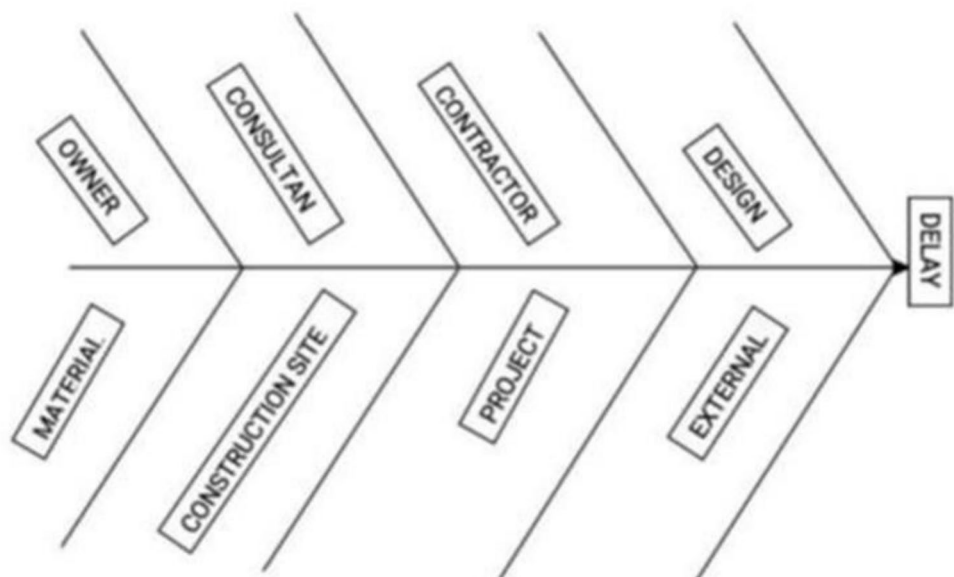


Table 2 RII and delay factor ranking

Factor group	Number	Factors causing delays	L	V	M	H	W	RII	Rank
Owner/client- representative related factors	1	Delay in decision making	10	31	27	4	0	0.469	28
	2	Delay in revising and approving documents	5	19	40	8	0	0.541	16
	3	Delay of financing and payments by owner	10	12	40	10	0	0.538	18
	4	Delay in delivery construction site to contractor	8	14	40	10	0	0.544	15
	5	Type of project bidding and award	31	28	9	4	0	0.361	50
	6	Unrealistic enforced contract duration	16	41	14	1	0	0.4	46
	7	Lack of experience of owner in construction projects	22	32	11	7	0	0.408	45
	8	Delay by owner in handling over process or approval of completed work	15	9	33	15	0	0.532	21
	9	Suspension of work	18	10	26	15	3	0.53	22
	10	Changes to the project by owner	20	33	11	8	0	0.419	42
Consultant representative related factors	1	Lack of experience of consultants	13	39	14	5	1	0.434	39
	2	Delay in approval of submittals design drawings and sample materials	6	16	41	9	0	0.547	13
	3	Mistakes or Discrepancies in documents or specification caused by consultants	15	37	16	4	0	0.425	40
	4	Poor communication and coordination with other parties	12	11	37	9	3	0.54	17
	5	Delay inspection	8	21	32	4	7	0.558	9
	6	Inadequate project management	18	27	22	5	0	0.438	37
Contractor related factors	1	Difficulties in financing the project by contractor	11	38	17	3	3	0.458	31
	2	Poor site management and supervision	13	31	20	7	1	0.466	29
	3	Deficiency in planning and scheduling	10	20	33	5	4	0.525	23
	4	Rework due to error during construction	8	9	38	9	8	0.6	5
	5	Delays related to subcontractors work	7	23	26	11	5	0.55	12
	6	Inappropriate construction methods	25	25	14	6	2	0.418	43
	7	Poor communication and co- ordination with other parties	8	17	35	10	2	0.546	14
	8	Unsafe practice at site	10	13	34	14	1	0.552	11
	9	Lack of experience of contractor	16	32	17	7	0	0.441	34
Factor group	Number	Factors causing delays	L	V	M	H	W	RII	Rank
Design related factors	1	Complexity of project design	8	38	22	3	1	0.463	30
	2	Mistakes and delays in producing design documents	11	32	28	1	0	0.452	32
	3	Poor use of advanced engineering design software	30	22	14	4	2	0.394	47
	4	Unclear and inadequate details in drawings	14	36	18	3	1	0.436	38
	5	Insufficient data collection and survey before design	15	36	20	0	1	0.422	41
	6	Design errors made by designers	8	15	44	3	2	0.533	20
Material related factors	1	Shortage of construction material	8	9	25	28	2	0.619	2
	2	Delays due to material delivery	3	12	23	31	3	0.652	1
	3	Change in material types and specifications during construction	9	22	33	7	1	0.513	24
	4	Inflation and escalation of material prices	12	24	26	10	0	0.494	25
	5	Unreliable suppliers	6	24	34	8	1	0.536	19

Table 2 (continued)

Factor group	Number	Factors causing delays	L	V	M	H	W	RII	Rank
Construction site related factors	1	Shortage of equipment failure	11	10	27	21	3	0.556	6
	2	Unforeseen site conditions	9	17	29	13	4	0.561	8
	3	Restriction at job site	7	15	26	18	6	0.602	4
	4	Lack of site utilities or services such as (water, electricity)	9	14	24	22	2	0.575	7
	5	Accident during construction	13	12	29	14	4	0.555	10
	6	Problem with nearby structure or facilities	21	4	19	3	5	0.451	33

L-Low, V-Very Low, H-High, W-Very high, M-Moderate

Table 3 Mean RII and ranking of groups of delay factors

Group of factors	Mean RII	Rank
Owner/client representative	0.472	5
Consultant	0.490	4
Contractor	0.506	3
Design	0.450	6
Material	0.562	1
Construction site	0.555	2
Project	0.422	8
External	0.429	7

study conducted a disaggregated examination by calculating separate Relative Importance Index (RII) scores for roads, bridges, and utilities projects.

Lastly, the study has conducted Factor Analysis of Delay Causes in Infrastructure Construction Projects. The study analyzed 25 delay causes identified from Table 1 and case studies, using survey responses collected from 72 construction professionals. The questionnaire employed a 5-point Likert scale (1 = very low to 5 = very high) to measure perceptions of each delay factor's impact. This dataset preparation ensured comprehensive coverage of potential delay causes while maintaining standardized measurement across all responses. The sample size of 72 professionals provided sufficient statistical power for subsequent factor analysis while representing diverse perspectives from the construction industry. All variables were carefully coded and checked for completeness before analysis to ensure data quality and reliability. Factor analysis was conducted using Principal Component Analysis (PCA) with Varimax rotation to extract meaningful factors.

Results

RII Factor

Based on the ranking of the groups, several significant elements were identified. These are discussed below:

Material (RII = 0.562)

One crucial factor contributing to delayed construction is the material-related group. This issue arises from problems such as delays in material delivery (RII = 0.652), shortages of construction material (RII = 0.619), and unreliable suppliers (RII = 0.536).

Construction site (RII = 0.555)

The construction site-related group emerges as the second significant factor contributing to construction delays, with an RII of 0.602. This issue encompasses challenges such as scarcity of equipment and lack of site utilities, both with high RIIs of 0.575.

Contractor (RII = 0.506)

Following the construction site-related group, the contractor-related group stands out as the third most significant delaying factor. Key factors within this group include rework due to errors (RII = 0.600), sub-contractor work (RII = 0.555), and unsafe practices at the site (RII = 0.552).

Consultant (RII = 0.490)

Ranked fourth in importance, the consultant-related group presents significant delaying factors in construction projects. Notable elements within this group include issues such as consultant responsiveness (RII = 0.558), delays in the

approval of submittal designs ($RII = 0.547$), and insufficient communication and cooperation among project stakeholders ($RII = 0.544$).

Owner/Client ($RII = 0.474$)

Ranked fifth among delaying factors is the owner/client-related group. This category encompasses significant issues such as delays in supplying materials to the construction site ($RII = 0.544$), revisions and approvals of documents ($RII = 0.541$), and delays in financing and payments by the owner ($RII = 0.538$).

Design ($RII = 0.450$)

Following the owner/client-related group, the design-related group emerges as the sixth significant delaying factor. Key factors within this group include design mistakes made by designers ($RII = 0.533$), challenges in project design ($RII = 0.463$), and errors and delays in providing design documents ($RII = 0.452$).

External ($RII = 0.429$)

Ranked seventh, the external-related group comprises significant factors contributing to delays. These include the impact of weather conditions ($RII = 0.616$), changes in government rules and regulations ($RII = 0.475$), and delays in third-party certification and final inspections ($RII = 0.350$).

Project ($RII = 0.422$)

The project-related group is identified as the last contributing factor to delays. Key elements within this group include project complexity ($RII = 0.472$), inadequate substantial completion ($RII = 0.441$), and ineffective delay penalties ($RII = 0.436$).

For this study, 15 of the most significant factors causing delays have been identified. Among these top factors, 33% are related to construction sites, 26% are related to contractors, 15% are related to materials, and 6% are related to external factors and owner/client issues. The questionnaire survey revealed these significant causes of delay.

Sector-Specific RII Factor

Table 4 presents the RII ranking for sector specific data. The analysis uncovered significant differences in critical delay factors—notably, land acquisition emerged as substantially more impactful for road projects ($RII = 0.68$) compared to bridge projects ($RII = 0.42$), while design errors showed the opposite pattern with higher importance for bridges ($RII = 0.55$) than roads ($RII = 0.30$). To statistically validate these

Table 4 Sector-specific RII rankings

Delay factor	Roads (RII)	Bridges (RII)	Utilities (RII)	<i>p</i> -value
Land acquisition	0.68	0.42	0.35	0.003
Design errors	0.3	0.55	0.48	0.012
Material shortages	0.62	0.58	0.6	0.21

observed variations, the study employed the Kruskal–Wallis non-parametric test, which confirmed significant differences ($p < 0.05$) in RII rankings for nearly half (12 out of 25) of the examined delay factors across sectors. These findings emphasize the necessity of tailored delay mitigation strategies that account for project-type specific vulnerabilities, particularly regarding land procurement challenges in road construction versus design precision requirements in bridge projects. To examine the cascading impacts between different delay factors, the study conducted Spearman's rank correlation analysis, which revealed several significant relationships. The analysis identified particularly strong correlations ($\rho > 0.5$) between key delay factors: late payments showed a substantial correlation with subsequent subcontractor delays ($\rho = 0.63$), which in turn were strongly associated with increased rework requirements ($\rho = 0.57$). Similarly, weather-related delays demonstrated a significant correlation with material shortages ($\rho = 0.61$), which were then linked to subsequent site-access problems ($\rho = 0.59$). These findings highlight how primary delays can trigger secondary and tertiary impacts throughout the project timeline, creating compounding effects that significantly extend overall project duration. The results emphasize the importance of addressing root-cause delays to prevent their downstream consequences on construction schedules.

Factor Analysis Approach

Factor analysis was conducted using Principal Component Analysis (PCA) with Varimax rotation to extract meaningful factors. The Kaiser criterion (eigenvalues > 1) was applied to determine the number of factors to retain, ensuring they collectively explained more than 60% of the total variance. A factor loading threshold of ± 0.5 was set to identify significant variable-factor relationships. The analysis demonstrated adequate sampling suitability with a Kaiser–Meyer–Olkin (KMO) measure of 0.78 (exceeding the 0.6 threshold) and Bartlett's test of sphericity confirmed significant correlations among variables ($p < 0.001$). Multicollinearity was assessed, with all variables showing Variance Inflation Factor (VIF) values below 5, indicating no problematic correlations between predictor variables. This methodological

approach ensured robust factor extraction while maintaining interpretability of the results.

Table 5 presents the results of the factor analysis. From the analysis, four factors were retained, explaining 68.3% of total variance. The factor analysis revealed four key dimensions influencing construction delays. First, Resource & Supply Chain (F1) emerged as the most significant factor, primarily driven by material-related issues, which corroborates the Relative Importance Index (RII) findings where material delays ranked highest. Practical mitigation strategies include establishing contracts with dependable suppliers and maintaining buffer stocks. Second, Government & External (F2) encompassed both administrative hurdles like permit approvals and land acquisition along with environmental factors, with case studies confirming land acquisition as a particularly critical challenge. Proactive engagement with regulatory bodies and implementing risk-sharing contractual agreements can help address these delays. Third, Contractor Management (F3) highlighted the impact of poor workmanship and supervision, consistent with the RII's third-ranked contractor-related factors. Enhancing workforce training and strengthening quality control measures represent effective countermeasures. Finally, Design & Planning (F4) uncovered fundamental pre-construction deficiencies, suggesting the need for adopting advanced Building Information Modeling (BIM) technologies and conducting thorough design reviews to prevent downstream delays. Together, these factors provide a comprehensive framework for understanding and mitigating delays in infrastructure projects.

Delay Mitigation Plan and Measures

From the questionnaire survey, significant causes of delay have been identified. To address these issues effectively, a delay mitigation plan has been developed. This plan aims to assist stakeholders during the execution stages of construction projects and is presented in Table 6. Reducing project delays involves evaluating mitigating steps during

the early phases of construction projects. It's essential to predict, identify, and implement cost-effective solutions. The following strategies can help stakeholders remain vigilant throughout project execution:

Key factors identified in the case study include local political conditions, obtaining permissions from local authorities, road diversions due to public conflicts, land acquisition challenges, material shortages, skilled labor scarcity, delayed payments, lack of government funds and dues, and public issues in construction areas. These factors need strategic addressing by stakeholders:

1. *Mitigating Delays due to Local Political Conditions*
The government should take steps to minimize political party influences on project activities. Collaboration among government institutions is essential to counter interference. Initiatives to limit the power of political leaders and local authorities are necessary.
2. *Obtaining Permissions from Local Authorities*
Pre-application discussions between developers and local authorities can streamline the permission process.
3. *Managing Road Diversions*
Effective communication strategies are crucial for informing the public about road works and temporary traffic diversions in advance.
4. *Addressing Land Acquisition Challenges*
Civil engineers should actively participate in property acquisition scheduling, particularly for government projects like road construction.

By implementing these strategies, stakeholders can mitigate delays and ensure smoother project execution. Addressing material shortages requires thorough research into material accessibility before construction commences. Similarly, labor requirements should be carefully assessed, and efforts should be made to retain trained personnel throughout the construction process. Employing experienced project managers can help mitigate client-related

Table 5 Results of factor analysis

Factor	Eigenvalue	% Variance explained (%)	Key variables (Loadings >0.5)
F1: Resource & supply chain	4.92	24.6	Material delivery delays (0.82) Shortage of construction materials (0.79) Unreliable suppliers (0.71)
F2: Government & external	3.45	17.2	Land acquisition (0.76) Obtaining permits (0.69) Weather effects (0.65)
F3: Contractor management	2.87	14.4	Poor site supervision (0.73) Rework due to errors (0.68) Inappropriate methods (0.61)
F4: Design & planning	2.44	12.1	Design errors (0.77) Insufficient data collection (0.64) Unclear drawings (0.58)

Table 6 Delay mitigation plan

S No	Causes of delay	Group	Mitigation plan
1	Slowdown in taking decision	O/C	Easy and Quick decision should be made
2	Changing and certifying documents	O/C	Owner should finalized the work priority before the execution
3	Delivery construction site to contractor	O/C	By paying extra bonus to the contractor and labours so as to complete work as per revised activities
4	Late payments by the owner	O/C	Prior payments to the vendors
5	Approval of completed work	O/C	Before execution, approving work should be fixed based on priority
6	Changes in the scheduling by owner at the time of construction	O/C	If any changes to be made the owner, it should inform to the contractor in advance
7	Delay in approval of submittals design drawings and sample materials	CS	If any alterations are done then Design engineer and consultant should seriously look into it. Arrange a meeting in case of any delay from design engineer and consultant for clearing doubts
8	Mistakes and discrepancies in documents	CS	All the Design documents should be assessed by consultant
9	Delay inspection	CS	Periodically the consultant should inspect the site for proper execution
10	Inadequate governance and supervision	Co	Well knowledgeable staff and workers should be recruited for effective site management
11	Deficiency in planning & scheduling	Co	It should be done by well qualified persons
12	Repetition of work	C	An Engineer or Architect should guide the ongoing project to avoid the repetition of work

O- Owner, C- Client, CS- Consultant, Co-Contractor

issues, which have emerged as significant factors in analyzing construction delays.

To prevent delays due to financial constraints, clients should ensure adequate financial resources are available before construction begins. Contractors should invest in the latest technology and machinery and provide training to staff to optimize efficiency and save time during project execution. Delayed payments are a significant cause of delays in road construction projects, often resulting from insufficient budgeting and funding prior to project awarding. Contractors should develop strategic transportation plans to mitigate delays caused by material shortages, ensuring timely delivery to the construction site. Governments should allocate budgets and award projects within set timeframes to address delays stemming from lack of funds and dues. Additionally, addressing public issues, such as land acquisition challenges, requires thorough verification of supporting documents before purchasing land to prevent future complications.

Conclusion

The study on delay analysis in Indian infrastructure construction projects, particularly focusing on roads and bridges, highlights the pervasive issue of project delays and their significant impact on time and cost. Through a comprehensive methodology involving questionnaire surveys and the Relative Importance Index (RII) analysis, the research identifies key delay factors and proposes mitigation strategies.

Key Findings:

1. Primary Causes of Delays:

1. Material-related issues (RII = 0.562) top the list, including delays in material delivery, shortages, and unreliable suppliers.
2. Construction site-related factors (RII = 0.555), such as equipment shortages and lack of utilities, rank second.
3. Contractor-related issues (RII = 0.506), like rework due to errors and subcontractor delays, are the third major cause.
4. Owner/client-related delays (RII = 0.474) and consultant-related delays (RII = 0.490) also significantly contribute to project slowdowns.

2. Sector-Specific Variations:

1. Road projects face significant delays due to land acquisition (RII = 0.68), while bridge projects are more affected by design errors (RII = 0.55).

3. Factor Analysis:

Four critical dimensions of delays were identified:

1. Resource & Supply Chain (material shortages, delivery delays).
2. Government & External Factors (land acquisition, permits, weather).

3. Contractor Management (poor supervision, rework).
4. Design & Planning (design errors, unclear drawings).

Mitigation Strategies:

1. *For Material Delays* Establish contracts with reliable suppliers and maintain buffer stocks.
2. *For Land Acquisition* Engage in proactive planning and fair value assessments to minimize conflicts.
3. *For Design Errors* Use advanced tools like BIM and conduct thorough design reviews.
4. *For Payment Delays* Ensure transparent payment procedures and adequate funding before project commencement.
5. *For Contractor Issues* Enhance workforce training and enforce strict quality control measures.

Time management is as critical as cost and quality in construction projects. Addressing root causes of delays through collaborative efforts among stakeholders can significantly improve project outcomes. This study provides a framework for understanding and mitigating delays, contributing to the timely and efficient execution of infrastructure projects in India. By implementing the proposed mitigation plans, stakeholders can reduce delays, enhance productivity, and contribute to the nation's infrastructure development goals.

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Declarations

Conflict of interest A competing interests declaration is mandatory for publication in this journal. Please confirm that this declaration is accurate, or provide an alternative.

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